6 Feature Effects Visualization

In this section, we introduce EffectVis, an interactive interface and tool built for exploring feature effects. On the top left panel, users can search for features and view their feature visualization charts – similar to Neuronpedia [Lin and Bloom, 2023]. Features can be added as feature cards to the canvas when you want to inspect them further.

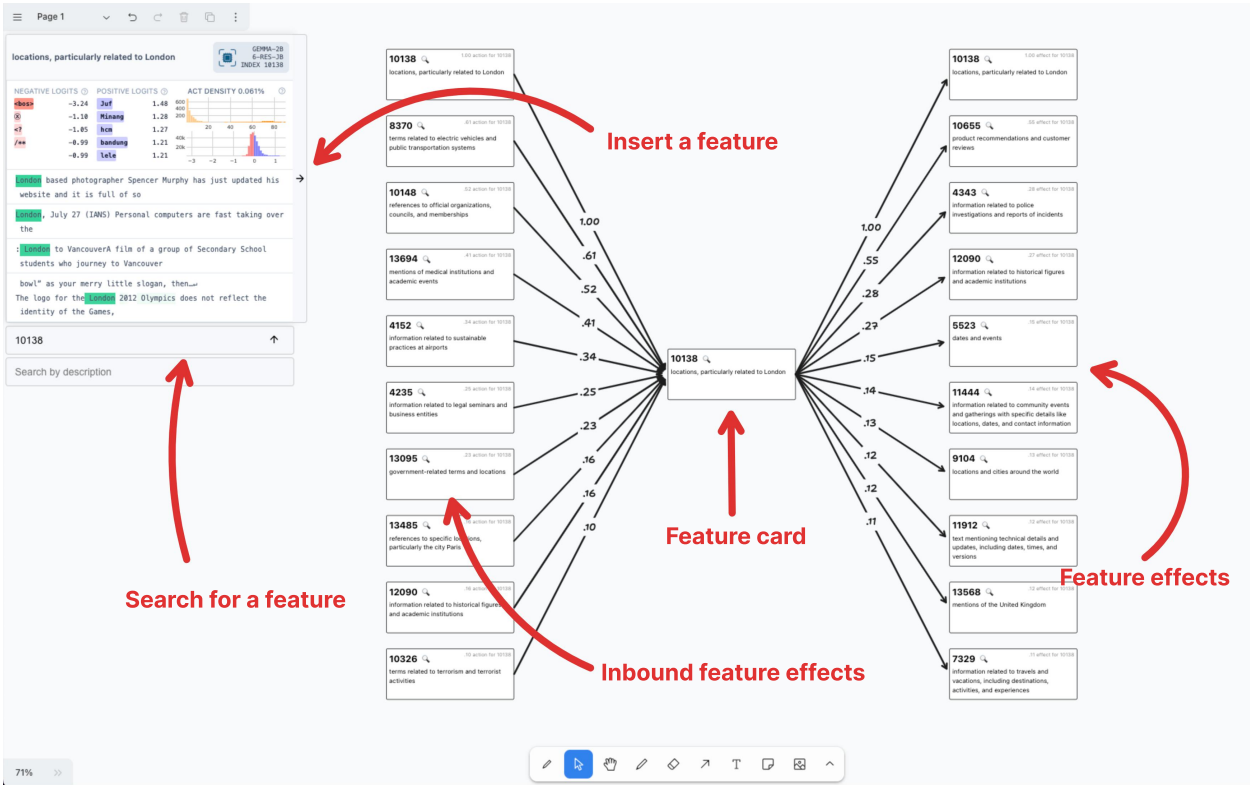

Figure 4: Interface overview for EffectVis. Available at https://effectvis.vercel.app/

For a selected card, our tool allows you to query for three different properties of the feature:

- The feature effects.
- The feature actions. These are the features that cause the selected feature to activate more often.
- Cosine similar feature directions.

In addition, when a feature card is selected, other cards for the same feature are highlighted, making it easy to see here else a feature is used at a glance. By clicking the magnifying icon on the top right of a feature card, the top panel will be populated with the relevant information for that feature.

Prior to developing EffectVis, we had a multi-step process for exploring and making sense a feature's effects and/or cosine similarity. First, we would access the list of effects for a feature in a python notebook. Then, using Neuronpedia, we would search for each feature one by one. These were then added to a list on Neuronpedia.

Due to the time-consuming nature of the process, we were limited in how many features we could inspect, reducing our ability to build intuition for what was being measured with the feature effects. EffectVis solved this for us by compressing the above steps into a single keyboard command action. It also made comparisons between feature effects and/or decoder cosine similarity easier. Lists of features could be pulled up side-by-side and compared visually. For comparing many lists, selecting a card shows you the other lists it is a part of.

EffectVis helped us develop the insights that led to our Optimized Steering Vectors method. In the future, it may useful to incorporate functionality for querying our feature effects by different filters such as effect scores as we did for part of our analysis in the feature effects section. This would allow us to perform more experiments and ask a wider variety of questions using the interface that we resorted to using python notebooks to explore. We haven't rigorously compared EffectVis to alternative tools, as it is tailored for the standalone method that we developed.

7 Discussion

7.1 Limitations

We performed all of our investigations using Gemma-2 2B and 9B, both of these models share the same distilled model architecture, so our results may not transfer to other models. Additionally, these are base models which haven't been instruction tuned, so we focused on steering in the open-ended generation setting and did not investigate tasks and features which would be relevant for safety or capabilities.

We focused only on steering towards concepts present in SAEs, which is a problem for multiple reasons. Firstly, some concepts don't appear as individual SAE features, a phenomenon called feature splitting. Secondly, niche features may only appear in extremely large SAEs, as discussed in the "feature completeness" section of Templeton et al. [2024]. These problems are particularly relevant because our targeted steering method, as described in Section 4, can only steer towards features present in the SAE used to measure feature effects. We explore a potential solution in Appendix D.

7.2 Future work

An important direction for future work is to apply our steering methods to chat models and safety-relevant steering targets to evaluate effectiveness in practical scenarios. Additionally, this work can be extended by exploring different SAE architectures, such as TopK SAEs [Gao et al., 2024], and testing on various model architectures beyond the Gemma family. Another direction to explore is adapting BiPO [Cao et al., 2024] to optimize steering vectors towards desired feature effects.

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